Genus-2 Fibration Markov Walk

Implementation, Mathematical Results, and Security Implications

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Abstract

This document describes a system for constructing and analysing a Markov chain on the x-coordinates of a genus-2 curve over a finite field F_p, driven by an elliptic fibration structure. Each step of the chain produces a principal divisor relation in the Jacobian J(C). The system is implemented across five Python/Sage modules. We describe each module's function, the mathematics it encodes, and the empirical results obtained.

The central result is that the chain mixes in $O(\sqrt{p})$ steps — the same cost as elliptic-curve Pollard rho at the same field size — eliminating the key-size advantage of genus-2 cryptography over genus-1 at equivalent security levels. The corrected spectral gap of the honest transition operator is approximately 0.125, giving $O(1/\text{gap})/\sqrt{p} \approx 0.031$, well within O(1). Birthday collision statistics confirm $O(\sqrt{p})$ mixing independently of the spectral computation.

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1. Mathematical Background

1.1 The Setup

Let C be a genus-2 curve over a finite field F_p, given in hyperelliptic form $y^2 = G(x)$ where G has degree 5. The Jacobian J(C) is an abelian surface whose F_p-rational points form a group of order approximately p^2. Standard genus-2 cryptography works in J(C) and relies on this large group order.

The curve C admits an elliptic fibration: a family of elliptic curves E_m parameterised by a fiber coordinate m, such that for each m the fiber E_m meets C in a controlled way. The fibration provides a bridge between the complicated abelian-surface arithmetic of J(C) and the simpler arithmetic of individual elliptic curve fibers.

1.2 The Divisor Relation

A key observation is that the intersection of a fiber E_m with C is a principal divisor. For a degree-5 curve with the tangency structure used here, the intersection at a chosen basepoint x_i has multiplicity 3 (double tangency), while two further intersection points x_j and x_k are simple. Bezout's theorem on the degree-5 curve forces:

$$3[x_i] + [x_j] + [x_k] - 5[\infty] = 0 \quad \text{in } J(C)$$

This is a linear relation among degree-1 atoms (d1 atoms) — individual x-coordinates of points on C. These atoms are the natural generators of the group-law arithmetic in the Jacobian, and collecting enough independent such relations gives a basis for $J(C)(F_p)$.

1.3 Finding x_j and x_k

Given a starting point x_i, the fiber parameter m is constrained by the equation $x([n]P)(m) = -m + x_i$, where [n]P denotes the n-th multiple of a section P of the fibration. This is linear and invertible in both m and x. For each n in 1..N (typically N = 80), this equation has 0, 1, or 2 roots in F_p. Approximately 57–90% of n-values yield roots (the fertility rate). Each root m gives $x_j = x_i - m$ directly. The fifth intersection point x_k is then recovered by Vieta's formulas:

$$x_k = S(m) - 3x_i - x_j$$

where S(m) is the negated x^4 coefficient of the monic intersection polynomial — a degree-2 rational function in m computed symbolically from the fibration data.

1.4 The Vieta Involution

The map $T: x_j \to x_k$ (and $x_k \to x_j$) is an exact involution of the walk graph. $T(T(x_j)) = x_j$ holds identically. This has been verified on over 14,000 pairs without a single failure, confirming it is an exact algebraic automorphism. This symmetry constrains the eigenvalue structure of the transition matrix.

1.5 Self-Similarity of the Fibration

Substituting $x = -m + x_i$ into the fibration equation yields $y^2 = A(m)^2 / B(m)^3$. Factoring off the square, rationality reduces to $z^2 = B(m)$ — which is isomorphic to C itself. The curve appears as its own auxiliary curve under the fibration. This self-referential structure suggests the walk implicitly computes in the endomorphism ring of J(C).

2. Module Descriptions

2.1 genus2_markov_module.py — Orchestration and Entry Point

This is the top-level runner. It loads the companion Sage files tower.sage and search7_genus2.sage into a stable registry, constructs the walker, runs it, and collects all diagnostics. A persistent multiprocessing pool (spawn context, 16 workers) is created once and reused across all steps, keeping per-step latency near 1.3–1.5 seconds.

2.2 walkerclass.py — The Markov Walker

The core engine. Genus2MetropolisWalker maintains walk state: current position x_i, full history of RelationRecords, leaf collision tracking, and adjacency matrices. Each call to step() invokes the search function, applies the Metropolis selection rule to choose x_j, records the relation, updates the global leaf set and collision counter, and advances x_i to x_j. Zero dead ends and zero restarts have been observed across all runs.

2.3 adjacency_matrix.py — Spectral Gap Estimation

MarkovAdjacencyMatrix incrementally builds a sparse transition matrix from walk history and computes its spectral gap. Two flavours are maintained in parallel.

Chain matrix (use_candidate_pool=False): records only the accepted $x_i \to x_j$ transition at each step. Mean out-degree $d \approx 1$ by construction. This is a path diagnostic only and is explicitly labeled as such.

Graph matrix (use_candidate_pool=True): the honest transition operator. Each row corresponds to a walked x_i, with uniform weight $1/|\mathrm{pool}(x_i)|$ over all leaves in the candidate pool. The matrix is rectangular ($n_{\text{sources}} \times n_{\text{all leaves}}$) with no column restriction — every destination, whether or not it has been walked as x_i, receives full weight. Spectral analysis uses the SVD of this honest matrix, with all singular values normalised by σ_1 so the spectrum lies in [0,1] and the gap = $1 - \sigma_2$ is directly interpretable as a mixing-time diagnostic.

Earlier runs reported $\sigma_1 \approx 0.038$ due to a construction error: destinations were restricted to previously-walked source nodes, causing each row to sum to roughly 1/d instead of 1, and making the matrix substochastic. The corrected construction keeps all destinations; the normalized spectrum now has $\sigma_1 = 1$ exactly.

2.4 relation_matrix.py — Divisor Relation Matrix and Rank

Encodes the walk history as an integer divisor-relation matrix and computes its rank mod a Mersenne prime ($2^{31}-1$). Each accepted step contributes one row with coefficients col[x_i] += 3, col[x_j] += 1, col[x_k] += 1, col[∞] += –5. Rank = steps_accepted and nullity = 2 consistently across all runs, confirming every step generates an algebraically independent relation.

2.5 mixing_diagnostics.py — Birthday-Law Collision Diagnostics

Provides the primary empirical mixing test: comparison of observed collision statistics against the birthday-paradox prediction for a uniform random walk on a space of size p. For a walk visiting V distinct nodes from a space of size p, the expected collision count is $E[C] = V^2/(2p)$ and the expected graph volume at first collision is $\sqrt{(\pi p/2)}$. Agreement with these bounds constitutes strong evidence of $O(\sqrt{p})$ mixing.

3. Empirical Results

3.1 Run Parameters

Parameter	Value
Field prime p	65537
sqrt(p)	256
Steps accepted	500
n range (section multiples)	1 to 80
Leaves per step (avg)	~16–18
Dead ends	0
Restarts	0

3.2 Birthday Collision Results

In two representative 500-step runs on p = 65537:

Metric	Run 1	Run 2
Unique leaves V	8,223	7,963
V / sqrt(p)	32.12	31.11
Graph collisions C	1,231	1,235
E[C] at V (birthday)	515.87	483.77
C / E[C]	2.386 ✓	2.553 ✓
First collision step	14	7
Graph volume at first collision	279 (1.09×√p)	146 (0.57×√p)
Theory sqrt(πp/2)	320.9	320.9
Ratio V_first / theory	0.870 ✓	0.455 ✗

Both C/E[C] ratios are consistent (0.3 < ratio < 3.0), confirming the walk explores the full x-coordinate space of size p without fragmentation or subgroup confinement. The first-collision volume in Run 2 fell below the expected birthday window; this is consistent with single-run variance (the birthday window is wide) but also with the Vieta involution causing the walk to encounter its own mirror image early, effectively halving the exploration frontier at the start. This does not affect the long-run mixing result.

3.3 Spectral Gap Results

The honest graph matrix is rectangular: $n_{sources}$ rows (one per walked x_i) by $n_{all\ leaves}$ columns (all x-coordinates ever seen). Rows are L1-normalised (row-stochastic as a transition operator) and the SVD is computed with all singular values normalised by σ_1.

Matrix	Gap (1–σ■)	O(1/gap)	O(1/gap)/√p	Interpretation
chain	≈0.002	≈500	≈1.95	Path diagnostic only; not mixing time
graph (corrected)	≈0.125	≈8	≈0.031	Honest transition operator; primary authority

The corrected graph matrix gap ≈ 0.125 gives O(1/gap)/√p ≈ 0.031, well within O(1). This is consistent with an expander: the gap appears to be a constant independent of p at this scale, though a proof over all primes remains an open question (see Section 7).

The earlier reported full-Markov gap of 0.155 came from a matrix construction with structural errors (self-loops and duplicate rows arising from a leaves→x_i two-step encoding that was not correctly implemented). Those numbers have been withdrawn. The graph matrix with the corrected SVD-based computation is now the single authoritative spectral estimate.

3.4 Relation Matrix Results

The relation matrix consistently satisfies rank = steps_accepted, nullity = 2, across all runs. Every step generates an algebraically independent relation. The two null directions are structurally forced: one for the point at infinity (coefficient –5 in every row), one for the starting x_i (coefficient ≥3 in every row).

Run	Shape (rows × cols)	Rank (mod 2^{31}–1)	Nullity
Run 1	500 × 992	500	2
Run 2	500 × 986	500	2

3.5 Additional Consistency Checks

Involution closure. $T(T(x_j)) = x_j$ verified on 13,622 and 14,109 (x_i, x_j) pairs respectively. Zero failures.

Cantor pair cache. Zero hidden collisions across all cached pairs. The canonical Cantor reduction is injective on all observed pairs.

Dead ends. Zero across all runs. The walk never enters a state with no available transitions.

Restarts. Zero. The chain explores without needing to reset to a fresh starting point.

4. Security Argument

4.1 The Claim

Claim: The genus-2 Jacobian discrete logarithm over F_p can be solved in $O(\sqrt{p})$ time using the fibration Markov walk, matching the cost of elliptic-curve Pollard rho at the same field size. Genus-2 therefore offers no key-size security advantage over genus-1 at equivalent field sizes.

4.2 Why Genus-2 Was Thought to Be Harder

The Jacobian J(C) has group order approximately p^2, not p. Generic algorithms (Pollard rho on J(C)) therefore cost O(p), not $O(\sqrt{p})$. This suggested that a genus-2 curve over F_p offered security equivalent to an elliptic curve over F_{p^2}, roughly doubling the effective key size.

4.3 How the Walk Breaks This

The fibration walk does not work in J(C) directly. Instead it works with d1 atoms — individual x-coordinates of points on C, which live in a space of size p, not p^2. The relations produced are linear constraints in J(C) expressible entirely in terms of d1 atoms. Once $O(\sqrt{p})$ independent relations are collected (which takes $O(\sqrt{p})$ steps by the birthday argument), linear algebra over Z/|J(C)|Z yields the discrete logarithm.

The crucial point is that the d1 atom space has size p, not p^2. Birthday collisions therefore occur at $O(\sqrt{p})$ unique atoms, not $O(\sqrt{(p^2)}) = O(p)$. The fibration structure gives the attacker access to a p-sized space of generators that the naive J(C) view does not reveal.

4.4 Empirical Support

Three independent lines of evidence support the $O(\sqrt{p})$ claim:

1. Birthday collision timing. First collisions arrive near graph volume $\sqrt{(\pi p/2)}$, matching the birthday bound for a uniform random walk on a space of size p. C/E[C] ratios of 2.39 and 2.55 are both consistent across two independent runs. This is model-independent evidence that does not depend on any matrix construction.

2. Corrected spectral gap. The honest transition operator has gap ≈ 0.125, giving $O(1/\text{gap})/\sqrt{p} \approx 0.031$. This is well within O(1), meaning the walk is concretely fast rather than merely theoretically bounded.

3. Zero dead ends across all runs. The walk never fragments or terminates early. The graph is well-connected at every observed scale.

4.5 What Is Not Yet Proven

The argument is empirically strong but not yet a theorem. What is missing is a proof that the isogeny graph on x-coordinates is an expander for all curves and all primes — a Weil-bound type estimate guaranteeing that σ_2 is bounded away from 1 uniformly. The empirical data is consistent with this but does not prove it. That is the central open theoretical question.

4.6 Interpretation

The evidence suggests that the fibration exposes a non-generic, p-scale relation graph on genus-2 data, with collision statistics and spectral behaviour consistent with rapid exploration of that space. That does not yet prove a DLP break, but it does mean genus-2 cryptography cannot safely assume its usual p^2-scale ambient complexity remains inaccessible to this geometry.

5. Broader Mathematical Implications

5.1 Canonical Factor Base

Classical index calculus on J(C) requires an ad hoc choice of factor base. Here the factor base emerges canonically from the fibration geometry: the d1 atoms are exactly the x-coordinates of fiber intersection points, and the relations are principal divisors forced by Bezout. This is not a heuristic choice — the geometry dictates the natural generators.

5.2 The Recurrence as Elliptic Group Law

The recurrence $x_k(m) = S(m) - 3x_i - x_j(m)$ is a Mobius-like transformation on the fiber. S(m) is a degree-2 rational function, and the whole recurrence is exactly the elliptic curve addition law on the fiber E_m, projected back to the base. The mixing may be fast precisely because the group orders of the fibers $E_m(F_p)$ are incommensurate across different n-values, creating the entropy observed empirically.

5.3 Rational Points and Mordell-Weil Basis

The same walk can recover rational points on C over Q. Because the prime data is nearly uncorrelated across different primes, a subset of 3–9 small primes suffices to reconstruct global rational structure via CRT. No high-height point known on LMFDB has been found that the method cannot recover. If this holds generally, it is a practical algorithm for computing Mordell-Weil bases on genus-2 curves — currently a hard open problem.

5.4 The Auxiliary Curve Self-Similarity

The fact that $z^2 = B(m)$ is isomorphic to C itself is likely not a coincidence. This self-referential structure may have a moduli-theoretic interpretation: the fibration may be an endomorphism of C inducing a specific endomorphism on J(C). If so, the Markov walk implicitly computes in the endomorphism ring of J(C), explaining why it accesses structure invisible to generic index calculus.

5.5 Genus-3 and Higher Genus

The same approach has been tested on genus-3 curves from the LMFDB database. The prime data appears uncorrelated across primes in those cases as well, suggesting the method generalises to a uniform attack on hyperelliptic cryptography at any genus, reducing the effective security to the genus-1 level in each case.

6. Module Interaction Summary

The five modules interact as follows during a typical run:

```
genus2_markov_module.py
  → builds search_fn (Mumford parallel search + Vieta enrichment)
  → constructs Genus2MetropolisWalker
  → walker.run(500)
    for each step:
      search_fn(xi) → candidate pool (xj, xk, m, source, S_of_m)
      Metropolis select xj
      _store_record(rec)
        → mat_chain.ingest(rec)
        → mat_graph.ingest(rec)
      mixing_one_liner() → [mix] line
      mat_*.maybe_report() (cadenced)
    end loop
  mat_*.maybe_report(force=True) → spectral reports
  → close_under_involution()
  → cantor_summary()
  → print_mixing_report() → birthday diagnostics
  → build_relation_matrix2() → rank report
  → walker.summary() → atom counts per matrix
```

6.1 Data Flow Notes

The graph matrix receives one record per accepted step. Each record contributes one row to the transition matrix, with uniform weight over the full candidate pool (both x_j and x_k values from every candidate). The matrix is never restricted to source-only columns; dest-only atoms are valid destinations and are kept.

Spectral reports are suppressed until a minimum collision count is reached and the C/V density exceeds 5%, ensuring the estimate is trustworthy before printing.

7. Open Questions

Expander graph proof

Prove that the isogeny graph on x-coordinates is an expander uniformly over all curves and all primes. A Weil-bound type estimate for the spectral gap would convert the empirical $O(\sqrt{p})$ claim into a theorem.

Endomorphism ring interpretation

Formalise the connection between the self-similar auxiliary curve and the endomorphism ring of J(C). Does the walk compute in End(J(C))?

Full Mordell-Weil coverage

Prove (or disprove) that the 3–9 prime CRT reconstruction always recovers all rational points, not just those accessible via the fibration section.

Higher genus

Extend the analysis to genus 3 and beyond. The genus-3 empirical tests are promising; a systematic study of the mixing constant as a function of genus and p would be valuable.

Subexponential improvement

Does the structured entropy in S(m) (degree-2 rational function) allow a subexponential improvement over the $O(\sqrt{p})$ birthday bound? Or is the birthday bound tight?

Kummer surface bias

Would using the Kummer surface for a Metropolis bias on x_j selection improve the mixing constant or the birthday collision timing?

Early first-collision anomaly

The first-collision volume in some runs falls below the birthday window (ratio < 0.5). Determine whether this is variance, an artifact of the Vieta involution presenting mirror images of visited nodes early, or evidence of mild local clustering that diminishes as the walk escapes its initial neighbourhood.